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Engineering cellulolytic bacterium *Clostridium thermocellum* to co-ferment cellulose- and hemicellulose-derived sugars simultaneously

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Abstract

Cellulose and hemicellulose are the most abundant components in plant biomass. A preferred Consolidated Bioprocessing (CBP) system is one which can directly convert both cellulose and hemicellulose into target products without adding the costly hydrolytic enzyme cocktail. In this work, the thermophilic, cellulolytic, and anaerobic bacterium, *Clostridium thermocellum* DSM 1313, was engineered to grow on xylose in addition to cellulose. Both *xylA* (encoding for xylose isomerase) and *xylB* (encoding for xylulokinase) genes from the thermophilic anaerobic bacterium *Thermoanaerobacter ethanolicus* were introduced to enable xylose utilization while still retaining its inherent ability to grow on 6-carbon substrates. Targeted integration of *xylAB* into *C. thermocellum* genome realized simultaneous fermentation of xylose with glucose, with cellobiose (glucose dimer), and with cellulose, respectively, without carbon catabolite repression. We also showed that the respective H₂ and ethanol production were twice as much when both xylose and cellulose were consumed simultaneously than when consuming cellulose alone. Moreover, the engineered xylose consumer can also utilize xylo-oligomers (with degree of polymerization of 2–7) in the presence of xylose. Isotopic tracer studies also revealed that the engineered xylose catabolism contributed to the production of ethanol from xylan which is a model hemicellulose in mixed sugar fermentation, demonstrating immense potential of this enhanced CBP strain in co-utilizing both cellulose and hemicellulose for the production of fuels and chemicals.

KEYWORDS

biohydrogen, *Clostridium thermocellum*, consolidated bioprocessing (CBP), lignocellulose, thermophile, xylose

1 | INTRODUCTION

Plant biomass (lignocellulose) is the most abundant renewable resource on earth, and its biological conversion to sustainable fuels

and chemicals has been a major research focus in the last 40 years (Lynd, Weimer, van Zyl, & Pretorius, 2002). Cellulose, hemicellulose, and lignin are three major constituents in lignocellulosic biomass, with the former two carbohydrates combined represent approximately 70%

of plant cell wall by weight (Lynd et al., 2002). Cellulose is a polymer of hexose sugars, and is difficult to hydrolyze due to its highly ordered crystalline structure (Himmel et al., 2007). Its association with hemicellulose and lignin further contributes to poor cellulase accessibility (Himmel et al., 2007). Currently the enzymatic hydrolysis of cellulose is a cost driver which prohibits the economic conversion of plant biomass to biofuels/chemicals in biorefinery (Bin Yang, Ding, & Wyman, 2011; Lynd et al., 2008). In contrast, hemicellulose is a pentose-rich polysaccharide that includes xylan and arabinoxylan, and relative to cellulose it can be easily hydrolyzed by dilute-acid pretreatment (Schell, Farmer, Newman, & McMillan, 2003). To overcome biomass recalcitrance, pre-treatments of the plant biomass with dilute base or acid (Chen et al., 2016) have been employed to increase the accessibility of both cellulose and hemicellulose to saccharolytic enzymes, and subsequently yield fermentable sugars. In this process scheme, non-cellulolytic microbes including *Saccharomyces cerevisiae* ferment the monomeric sugars liberated from enzymatic digestion to yield ethanol or products of interests via strain engineering (van Zyl, Lynd, den Haan, & McBride, 2007).

Naturally cellulolytic microorganisms have drawn increasing amount of attention as these microbes produce their own hydrolytic enzymes to breakdown cellulose, making direct fermentation of cellulosic biomass feasible. Such conceptually desirable process configuration to yield products in one integrated step without separate addition of exogenous enzymes is referred to as consolidated bioprocessing (CBP) (Lynd, Wyman, & Gerngross, 1999; van Walsum & Lynd, 1998). Several cellulolytic microbes are under characterization and strain development (Blumer-Schuette et al., 2011; Higashide, Li, Yang, & Liao, 2011; Kim, Chung, Himmel, Bomble, & Westpheling, 2017; Munir et al., 2016). Of these cellulolytic bacteria, *Clostridium thermocellum* is an anaerobe, producing an intricate multienzyme complex called the cellulosome (Bayer, Belaich, Shoham, & Lamed, 2004; Demain, Newcomb, & Wu, 2005; Doi & Kosugi, 2004; Schwarz, 2001), and displays one of the fastest growth rates on crystalline cellulose (Demain et al., 2005; Lynd et al., 2002). In addition, it is a thermophilic bacterium that grows optimally between 55 and 60 °C, and fermentation at elevated temperatures reduces contamination.

C. thermocellum breaks down cellulose to form cellobioses and cellobiose, the latter a hexose disaccharide. Yet this microbe cannot grow on hemicellulose (Demain et al., 2005). When referring to CBP involving *C. thermocellum*, previous work has thus focused extensively on the bacteria's ability to utilize six-carbon complex substrates, while a more ideal CBP configuration in a broader sense would entail CBP of all carbohydrates in plant biomass. To maximize the utilization of both cellulose and hemicellulose in plant biomass via CBP, co-cultures of cellulose-degrading "specialists" such as *C. thermocellum* with a separate pentose-consuming bacterium have been employed to degrade both six-carbon (C6) and five-carbon (C5) sugars (Argyros et al., 2011). However, enabling CBP using a single species has the advantages to bypass potential challenges arising from shifting species populations in a co-culture or consortium caused by changing substrates availabilities, pH, cellular redox, and end-products inhibition. *C. thermocellum* encodes several xylanolytic enzymes with their

expression accounting for 22% of the total spectral abundance in a proteomic analysis when *C. thermocellum* was cultured on cellobiose (Raman et al., 2009). Xylanase activity has been demonstrated in vitro (Morag, Bayer, & Lamed, 1990). However, it is unclear which enzymes are deficient in order for the bacteria to also utilize complex C5 substrates, and the cause for the lack of growth in xylan has been inferred to its inability to metabolize the resulting xylose or xylose oligomers. Overcoming the latter bottleneck is highly impactful and could potentially pave the way to afford next-generation CBP (NG-CBP) in a single strain effective in hydrolysing both cellulose and hemicellulose in the near term.

Our approach described herein is to introduce a non-native xylose-metabolizing pathway in *C. thermocellum* via genetic engineering. We characterize the utility of the engineered strain in consuming hexose, pentose, and mixtures of these sugars in different combinations. Because *C. thermocellum* naturally produces copious amounts of hydrogen (H₂) during fermentation (Carere, Kalia, Sparling, Cicek, & Levin, 2008; Lalaurette, Thammannagowda, Mohagheghi, Maness, & Logan, 2009; Magnusson, Cicek, Sparling, & Levin, 2009), and the renewable production of H₂ is of broad interest considering the trends forecasted to the global energy matrix (Meher Kotay & Das, 2008), we also compare the production of H₂ and ethanol during simultaneous co-fermentation of xylose and crystalline cellulose (Avicel) to that from fermentation of only Avicel. Such strain development in *C. thermocellum* provides a critical enabling step to realize the conceptually attractive NG-CBP approach using a single species.

2 | MATERIALS AND METHODS

2.1 | Plasmid and strain construction

C. thermocellum DSM1313-derived mutant Δhpt , also referred to as KJC315 (Xiong et al., 2016), was the parental strain in this study. The gene *hpt* (clo1313_2927) encodes for hypoxanthine phosphoribosyl transferase, which was used for counter-selection in the presence of 8-azahypoxanthine.

The *xylA* and *xylB* genes originated from the thermophilic anaerobic bacterium *Thermoanaerobacter ethanolicus* was driven by a relatively strong native promoter, glyceraldehyde 3-phosphate dehydrogenase promoter (PgapDH) (Olson et al., 2010). Individual genes and the promoter were PCR amplified, assembled, and inserted into the *PstI-SphI* sites of plasmid pXW15, generating pXW15-*xylAB* (Figure S1a, Supporting Information). To ensure stable protein expression, we chose to integrate the expression cassette in between the 3' and 5' flanking regions of the *hpt* gene via homologous recombination in the targeted *hpt* knockout host (KJC315) described previously (Xiong et al., 2016). Integration into this locus avoids any unknown interference which might arise elsewhere in the genome. Also on the plasmid (pXW15-*xylAB*) designed for markerless knock-in of *xylAB* genes contains the *tdk* gene encoding for thymidine kinase followed by the thiamphenicol resistance gene encoded by *cat* gene. CAT and TDK serve as selectable and counter-selectable markers,

respectively, which are driven by the native phosphoglycerate kinase promoter (Ppgk, Figure S1a, Supporting Information).

Transformation of the plasmid into the bacteria is as described previously (Olson & Lynd, 2012). Briefly, transformants were plated on CTFUD rich media containing 1% agar, 0.5% w/v cellobiose, and was selected with thiamphenicol (20 µg/ml) for cells carrying pXW15-xylAB.

Colonies grown on this plate were validated with PCR for the presence of plasmid pXW15-xylAB, revived in liquid media containing the same concentrations of cellobiose and thiamphenicol, and plated again on CTFUD medium supplemented with xylose (5 g/L) as the only added carbon and 5-Fluoro-2'-deoxyuridine (FUDR, 10 µg/ml). In this selection on plate, the native *C. thermocellum* strain cannot grow on xylose as the added carbon source, so presumably only the xylAB-expressing bacteria can grow on xylose. FUDR selects against cells expressing *tdk*, so the xylose/FUDR dual selection promotes losing the *tdk* on the plasmid hence curing the plasmid while promoting the integration of xylAB cassette into the chromosome through double homologous recombination. The correct genotype of the engineered strain named KJC335 was validated by several PCR reactions and commercial DNA sequencing. We also validated that KJC335 could no longer grow on thiamphenicol, indicating that the antibiotic marker and the plasmid backbone have been lost during counter-selection. All primer sequences used for colony PCR and constructing the plasmids are summarized in Table S1, Supporting Information.

2.2 | Culture conditions and medium

Unless otherwise noted, *C. thermocellum* DSM1313-derived strains were routinely grown anaerobically at 55 °C on 5 g/L (14.6 mM) cellobiose with initial media pH set at 7.0. Cellobiose was replaced by various carbon sources including xylose, xylo-oligomers, glucose, xylan, Avicel, or combinations of these substrates for testing the performance of sugar consumption in the engineered *C. thermocellum* strain. All chemicals were purchased from Sigma-Aldrich (St. Louis, MO) unless otherwise noted. We used the DSM122-rich medium referred to as CTFUD medium (Olson & Lynd, 2012). The growth medium was deoxygenated by gassing with argon gas for 20 min and autoclaved before use.

2.3 | Products measurement and sugar analysis

Cell growth was measured as a function of DCW and optical density by spectrophotometry (DU800; Beckman Coulter, Brea, CA) at OD₆₀₀. Cellobiose, xylose, formate, lactate, acetate, and ethanol were measured by HPLC (1200 series; Agilent Technologies, Santa Clara, CA) with a mobile phase of 4 mM sulfuric acid (flow rate of 0.6 ml/min) using an Aminex HPX-87H column with a Micro Guard Cation H Cartridge. The column temperature was set to 55 °C. H₂ in the headspace of sealed culture tubes was measured by injecting 200 µl headspace gas into GC (Agilent 6890 system equipped with J&W Mol Sieve 5A column and thermal conductivity detector). The temperature

and pressure of inlet were set as 150 °C and 25 psi, respectively. Column temperature was 60 °C with argon as the carrier gas.

To analyze xylo-oligomers, the samples were separated and detected using an HPLC system (1200 series; Agilent Technologies) equipped with an Evaporative Light Scattering Detector (ELSD). The chromatography system utilized a Shodex Sugar SZ5532 (Zinc) column (Showa Denko K.K., Tokyo, Japan) at 6 × 150 mm, 6 µm particle size, to sufficiently separate the oligomers of interest. The column temperature was maintained at 60 °C, and the solvents used to separate the analytes were water (A) and acetonitrile (B). The separation was carried out using a gradient program of (A) = 20% and (B) = 80% at time *t* = 0; (A) = 17% and (B) = 83% at *t* = 9 min; (A) = 30% and (B) = 70% at *t* = 25 min; (A) = 40% and (B) = 60% at *t* = 40 min; (A) = 20% and (B) = 80% at *t* = 45 min. The flow rate was held constant at 0.9 ml/min. The ELSD parameters were set so the drift tube temperature was set to 76 °C and the gain to 10.

To analyze the Avicel consumption by *C. thermocellum*, we digested all polysaccharides in CTFUD culture medium and quantified glucose as the digested product. Briefly, cultures were frozen evaporated and digested with 72 wt% sulfuric acid at 30 °C for 1 hr. The solute was diluted 10-fold and autoclaved at 121 °C for 1 hr, followed by neutralization with CaCO₃ (molar ratio to H₂SO₄ is 1:1) until no CO₂ was released. The mixture was centrifuged at 13,000g for 1 min and supernatant was filtered for glucose analysis using HPLC as described above.

2.4 | Isotope tracer analysis

KJC335 was grown in CTFUD medium with uniformly ¹³C-labeled glucose and/or unlabeled xylan from beechwood (Sigma-Aldrich). When OD₆₀₀ reached 1.5, cells were spun down and the isotope pattern of ethanol in supernatant was analyzed by GC-MS (Agilent GC-6890 gas chromatograph equipped with an Agilent 19091J-413 column (30 m × 0.32 mm × 0.25 µm) connected to a MS-5975C mass spectrometer). The inlet temperature was set at 225 °C. Oven temperature started at 40 °C for 3 min, followed immediately by a ramp of 45 °C/min to 235 °C and was held for 3 min. Injection of 1 µl sample with a split ratio of 25 in constant pressure mode (9.52 psi) at the inlet was applied.

3 | RESULTS

3.1 | Integrating a synthetic xylAB operon in *C. thermocellum* genome

C. thermocellum can grow on glucose, cellobiose, and cellulose but not on xylose. This is likely due to lacking in its genome genes encoding key xylose catabolizing enzymes such as xylose isomerase (XI) and xylulose kinase (XK). We selected transformants with thiamphenicol in cellobiose followed by counter-selection with 5-fluoro-2'-deoxyuridine (FUDR) in xylose for xylAB-integrated strain (KJC335) as illustrated in Figure S2, Supporting Information. A second synthetic operon was constructed and tested in parallel which drives the

expression of the same *xylAB* genes with cellobiose phosphorylase promoter (Pcbp). Only the synthetic operon driving the expression of *xylAB* using PgapDH and not Pcbp was able to grow on xylose and xylose/FUDR plate. We verified the genotype of this newly created strain isolated from the xylose and FUDR dual selection plate with several PCR reactions (Figure S1b, Supporting Information). In particular, the primer pair annealing to regions upstream of the *hpt* 5'-flanking and downstream of the *hpt* 3'-flanking sequences resulted in an expected DNA fragment (Figure S1b, Lane 7, Supporting Information), confirming that the entire *xylAB* cassette has been integrated into the target locus in genome. Commercial DNA sequencing has also validated the DNA sequence integrated at the target locus.

3.2 | *C. thermocellum* strain integrated with *xylAB* operon grows on xylose

To test whether KJC335 can metabolize xylose, we grew the engineered and its parental strains (Δhpt) on CTFUD rich medium in 3.5 g/L xylose as the carbon source without the addition of hexose sugars. At the beginning, KJC335 exhibited an approximately 20-hr lag period for xylose utilization, which was followed by rapid xylose consumption in the next 28 hr. In accordance with xylose consumption, an exponential growth was observed between 20 and 48 hr reaching ~ 1.2 g dry cell weight (DCW)/L at the end of fermentation (Figure 1a). In contrast to the KJC335 cultures, the non-engineered parental strain cannot grow on xylose. Xylose was not consumed, and its level remained unchanged over 70 hr (Figure 1a). These data showed that the *xylAB* was indeed expressed, and the expressed enzymes enable growth of *C. thermocellum* on xylose. We further compared growth and cellobiose utilization of these two strains in CTFUD medium. Results from Figure 1b showed that both KJC335 and Δhpt strains displayed closely comparable cellobiose consumption and growth phenotypes, demonstrating that KJC335 did not compromise its ability to grow on cellobiose comparing to the parental strain after the introduction of the non-native xylose metabolizing pathway.

3.3 | Engineered *C. thermocellum* strain co-utilizes xylose with gluco-monomer, -dimer, and -polymer

We further investigated the performance of *xylAB*-expressing strain on mixed sugar fermentation. KJC335 was first inoculated into CTFUD rich medium (Olson & Lynd, 2012) containing both 2.5 g/L xylose and 2.5 g/L glucose. As illustrated in Figure 2a, xylose and glucose were simultaneously utilized, with the continuous depletion of both sugars coincided with a rapid increase in cell biomass. Based on the slopes of sugar consumption, rate of xylose utilization (0.104 ± 0.010 g/hr/L) is faster than that of glucose (0.073 ± 0.005 g/hr/L). We next examined co-utilization of xylose with cellobiose, a glucose dimer produced during cellulose hydrolysis by *C. thermocellum*. We observed simultaneous consumption of xylose (at the rate of 0.037 ± 0.003 g/hr/L) and cellobiose with slightly faster cellobiose consumption rate (0.054 ± 0.007 g/hr/L, Figure 2b). We further tested co-fermentation

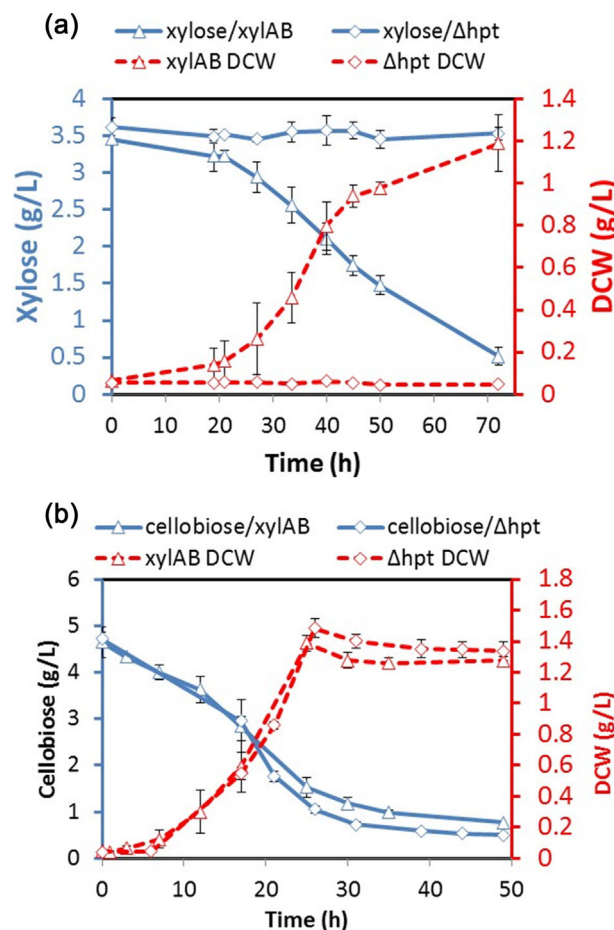


FIGURE 1 Growth and sugar consumption of *C. thermocellum* strain integrated with the *xylAB* expression cassette (triangle) and Δhpt (diamond) control on xylose (a) and cellobiose (b)

of xylose and Avicel® which is a model substrate of crystalline cellulose and polymer of glucose. Equal quantities of xylose and Avicel (2.5 g/L) were used as the substrate. To avoid interference from undissolved Avicel on cell biomass and optical density measurements, we quantified hydrogen production, a natural fermentation product in the headspace of sealed cultures as an indicator of fermentation. In theory, for every mole of glucose consumed 4 moles of H_2 can be produced maximally, and for every mole of xylose consumed 3.3 moles of H_2 can be produced maximally. The hydrogen produced from the combined feedstock of Avicel and xylose was about twice as much (16.7 ± 2.3 $\mu\text{mole/ml}$) as that from Avicel alone (7.9 ± 0.45 $\mu\text{mole/ml}$). Sugar consumption profiles also show that xylose was utilized simultaneously with Avicel. Unlike the case in xylose/glucose co-utilization, the depletion of Avicel (at the rate of 0.063 ± 0.001 g/hr/L) was faster than that for xylose (0.032 ± 0.005 g/hr/L). Consistent with H_2 production profile, ethanol is also produced twice as much during co-fermentation of Avicel and xylose than in Avicel alone (Figure 2d). Altogether, the series of experiments support that the common glucose effect also known as the carbon catabolite repression (CCR) (Gorke & Stulke, 2008), which impedes many microbes in mixed sugar fermentation does not exist between the combinations of

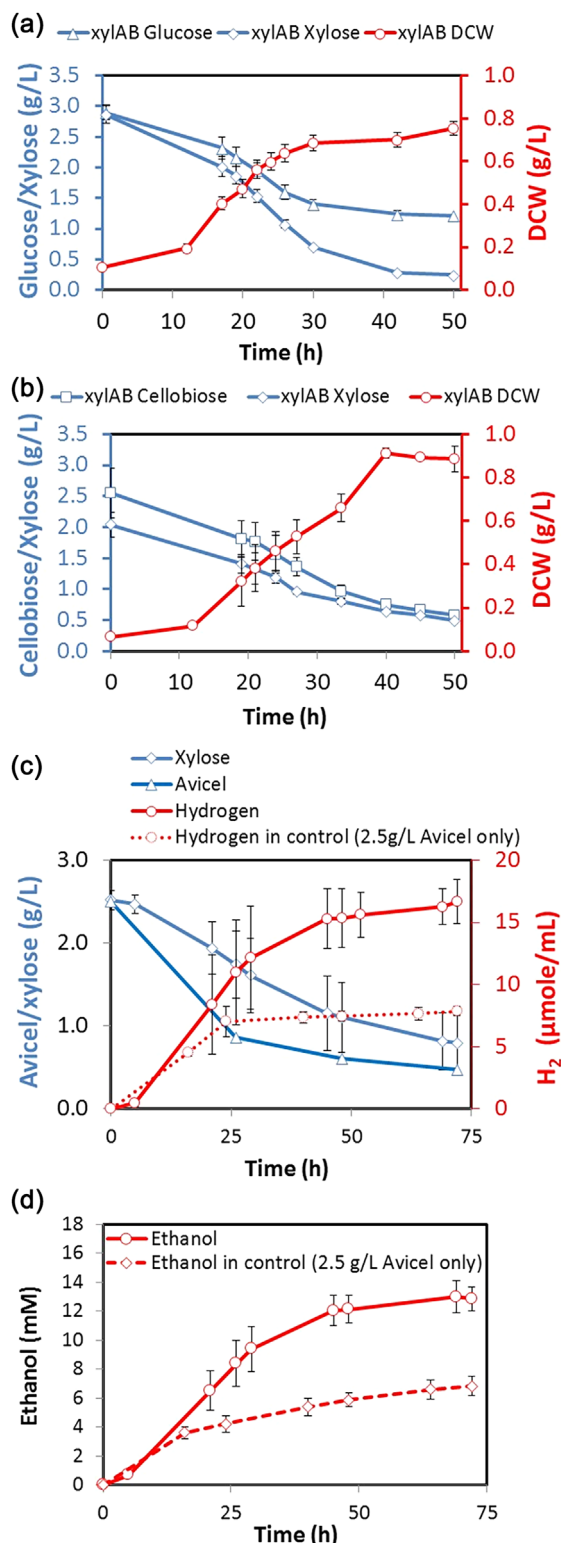


FIGURE 2 Simultaneous consumption of glucose/xylose (a), cellobiose/xylose (b), and Avicel/xylose (c) and the ethanol production over time with and without xylose during Avicel fermentation (d) by *C. thermocellum* xylAB integrated strain

mix-substrates tested in the engineered *C. thermocellum* strain. We observed simultaneous utilization of both pentose and hexose sugars in soluble or in complex forms without sequential consumption of the carbon sources. Data also support that the utilization of xylose

contributed to additional H₂ and ethanol produced during mixed C5/C6 co-fermentation.

3.4 | Xylose-engineered strain utilizes xylo-oligomers

C. thermocellum has the natural ability to breakdown xylan and exhibits endoxylanase activity by converting longer-chain xylose oligomers into xylose trimer and dimer (Morag et al., 1990), but without subsequent utilization of the resulting pentose sugars. This prompts the question if KJC335 can consume xylo-oligomers by coupling its inherent hemi-cellulolytic activity with the engineered xylose catabolic ability. We thus incubated KJC335 with 3.5 g/L xylo-oligomers (Degree of Polymerization: DP 2–7) in CTFUD medium. Results showed that cell biomass increased minimally during the first 20 hr from 0.05 to 0.12 g(DCW)/L (Figure 3a) and then ceased. However, when xylose (1.5 g/L) was added as the sole carbon substrate, cell biomass can reach up to 0.23 g(DCW)/L (Figure 3a), indicating the existence of a bottleneck for the utilization of xylo-oligomers alone. We then analyzed the changes in the levels of xylose tri-, di-, and monomers in the supernatant of these cultures. Xylose

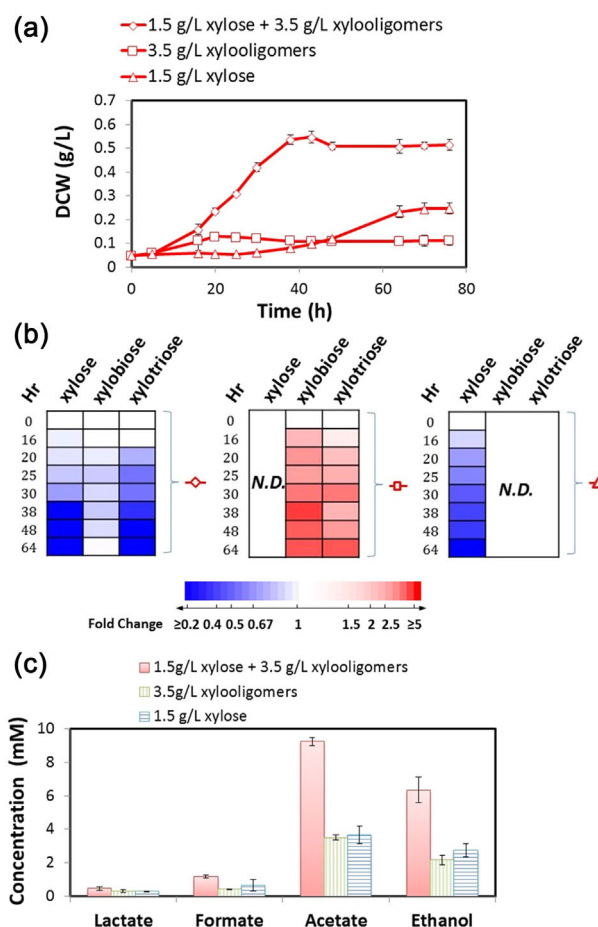


FIGURE 3 Kinetics of cell growth (a) and sugar utilization (b) and organic acids production (c) for xylAB expressing strain incubated with xylose (1.5 g/L) + xylo-oligomers (DP 2–7, 3.5 g/L) (open diamond), xylo-oligomers (DP 2–7, 3.5 g/L) (open square), and xylose only (1.5 g/L) (open triangle). ND, not detectable

dimers and trimers were accumulated to ~2.5-fold of their original level after 64 hr in cultures fed with 3.5 g/L xylo-oligomer (Figure 3b), suggesting the presence of certain xylanase activity which cleaves the xylo-oligomers with higher DP to xylotriose and xylobiose over time. Xylose monomer was not detected in the supernatant during fermentation, suggesting that further hydrolysis of the xylo- trimers and dimers into xylo-monomer was likely a rate-limiting step for KJC335 in utilizing xylo-oligomers.

The release of xylo-monomer from xylo-oligomers presumably relies on xylosidase activity which might be low when cultured in xylo-oligomers alone (Morag et al., 1990). To test the effect of xylose on xylobiose and xylotriose consumption, we fed 1.5 g/L xylo-monomer along with 3.5 g/L xylo-oligomers to KJC335 and examined the utilization of xylo-oligomers while xylo-monomer is not limiting at the beginning. As shown in Figure 3a, KJC335 indeed grew substantially with cell biomass reaching 0.53 g (DCW)/L at 40 hr. In comparison, cell growth on 1.5 g/L xylose can only reach ~0.23 g(DCW)/L after 72 hr, implying that additional cell growth was attributed from xylo-oligomers utilization. In parallel, we monitored the kinetics of xylo-oligomers level in xylose plus xylo-oligomers cultures. Both xylobiose and xylotriose level decreased in the supernatant, further supporting the utilization of xylo-oligomers by KJC335. Consistent with the substrates utilization pattern, major metabolic products including ethanol, acetate, and formate were produced in greater amounts in the combined feeding of xylose plus xylo-oligomers than in either xylose or xylo-oligomers alone at the end of the fermentation (Figure 3c). These findings collectively suggest that *C. thermocellum* expressing *xylAB* is able to consume xylo-oligomers conditionally, provided low amount of xylose is present initially.

3.5 | Engineered pathway enables xylan-to-ethanol conversion

To investigate the potential of the *xylAB*-expressing strain in converting lignocellulose-derived pentose to useful product, we tested its ability to convert xylan to ethanol during fermentation with an isotope tracer experiment. We fed KJC335 with 5 g/L unlabeled glucose, 5 g/L ^{13}C fully labeled glucose, or 2.5 g/L unlabeled xylan plus 2.5 g/L ^{13}C -glucose, respectively (Figure 4). It can be deduced that when feeding an unlabeled carbon source, the ethanol produced by KJC335 should contain only ^{12}C (Figure 4, row 1). Exclusive assimilation of ^{13}C -glucose will result in the incorporations of ^{13}C atoms in both carbons in ethanol (Figure 4, row 2). When ^{13}C -glucose was coupled to ^{12}C -xylan utilization, a mixture of unlabeled, partially labeled, and uniformly-labeled ethanol will be generated correspondingly (Figure 4, row 3). Therefore, the contribution of xylan to ethanol production can be manifested in ethanol labeling pattern which could be analyzed by mass spectrometry (MS). As shown, MS revealed two dominant m/z peaks at 45 and 46, respectively, for unlabeled ethanol (Figure 4, row 1, column 4) when cultured in unlabeled glucose. These peaks shifted to 47 and 48 with both carbons in ethanol ^{13}C -labeled upon feeding ^{13}C -labeled glucose (Figure 4, row 2, column 4). By feeding ^{13}C -glucose and unlabeled xylan, we detected

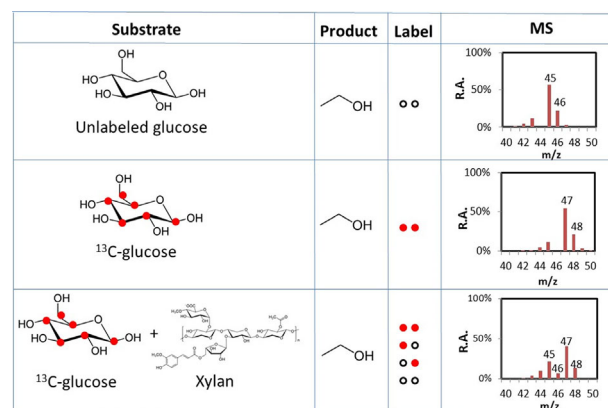


FIGURE 4 ^{13}C tracing of ethanol from unlabeled glucose (row 1), ^{13}C -glucose (row 2), and ^{13}C -glucose with unlabeled xylan (row 3) in *C. thermocellum* carrying *xylAB*. ^{13}C carbons are represented by red circles and ^{12}C carbons are represented by black circles

an array of isotope ions from 45 to 48 in which ions 45 and 46 indicated ethanol produced from unlabeled xylan. According to the relative intensity of fully unlabeled ethanol peaks to fully labeled ones (45/47 or 46/48 ratios), we estimated that approximately 30% of ethanol was produced from xylan in mixed-sugar fermentation. These results provide direct evidence that engineered pentose utilization in *C. thermocellum* indeed contributed to the production of biofuel in mixed sugar fermentation.

4 | DISCUSSION

Co-fermentation of both cellulose and hemicellulose from plant biomass without cross-inhibition can greatly improve biomass utilization, lower the feedstock cost, and enable the economic production of biofuels and biochemicals. Toward achieving this goal, we have successfully engineered the efficient cellulose-degrader *C. thermocellum* to also consume xylose. Characterizations of the engineered strain on substrate co-consumption, cell growth, H_2 and major organic acids production showed that the engineered strain can co-utilize xylose with six-carbon sugars including glucose, cellobiose, and Avicel, with Avicel being modeled crystalline cellulose. Moreover, the engineered strain is also capable of converting xylan to ethanol in the presence of glucose, based on altered ^{13}C -labeling patterns of the ethanol molecule when both substrates are present vs. with either substrate alone.

A major impediment in engineering microbe to co-utilize both pentose and hexose sugars efficiently is the glucose repression effect, also known as CCR, with utilization of glucose often preceding that of other sugars in mixed sugar fermentation (Brückner & Titgemeyer, 2002; Gorke & Stulke, 2008). CCR leads to longer fermentation and adds cost. It is therefore no surprise the microbial co-consumption of soluble pentose and hexose sugars without CCR has been earnestly sought after. The co-fermentation of xylose and cellobiose without the CCR effect has been reported in engineered model mesophilic organisms including yeast (Ha et al., 2011) and *E. coli* (Vinuselvi &

Lee, 2012). However, neither organism is engineered to be cellulolytic thus do not have the advantages of CBP. Recent advancement in engineering yeast to exhibit cellulolytic activities is in strains that cannot ferment pentose sugar (Liu et al., 2016). Besides CCR in mesophiles, CCR mechanism in the thermophilic and non-cellulolytic organism, *Thermoanaerobacterium saccharolyticum*, has been studied and its deregulation resulted in altered order of sugar utilization (Tsakraklides, Shaw, Miller, Hogsett, & Herring, 2012). There was one patent application filed on engineering *C. thermocellum* to co-ferment C5 and C6 sugars (Argyros, Barrett, Caiazza, & Hogsett, 2012). However, such strain's ability to simultaneously utilize both xylose with glucose, cellobiose, or crystalline cellulose was not shown or known. The strain presented herein was engineered with a distinct source of xylose-catabolizing enzymes and genetic approach, and provides the first substantial evidence that biomass-relevant complex substrate such as cellulose can be co-utilized with xylose simultaneously without CCR.

Five ATP-binding cassette (ABC) transporters in *C. thermocellum* have been identified and characterized previously (Nataf et al., 2009). Without engineering a transporter specific for xylose in the *xylAB* integrated strain, it is probable that xylose uptake in *C. thermocellum* was facilitated by native transporters promiscuous to which solutes it can bind.

Although simultaneous consumptions of xylose with glucose, cellobiose, and cellulose were observed, the differences in substrate consumption rates relative to that of xylose may reveal the microbe's efficiencies in sugar uptake and the subsequent ability to metabolize the sugar. Analyses on the bioenergetics of *C. thermocellum* growth on glucose and glucose-oligomers have shown that the utilization of long celldextrins provides more ATP per glucose molecule released than that obtained from cellobiose or glucose (Zhang & Lynd, 2005). Thus the cost of transport per hexose is higher for glucose than for oligosaccharides, and the uptake of longer oligosaccharides are energetically more favorable. It appears that mixed-sugar fermentation involving monomeric and oligomeric sugars can also be governed by this bioenergetics principle. In our case, the uptake of cellobiose and longer celldextrins followed by intracellular phosphorylytic cleavage would result in greater ATP gain than assimilating the monomeric glucose or xylose. Perhaps the bacteria have evolved to consume oligosaccharides faster than monomeric sugar to maximize the energetic gains as shown by our data (Figure 2). Indeed, *C. thermocellum* does not naturally grow on glucose with glucose utilization requiring adaptation (Demain et al., 2005; Lynd et al., 2002). The observed differences in relative sugar consumption rates could thus be a manifestation of a combination of the cells' underlying bioenergetics as well as the differential binding affinity between the sugar and the specific solute binding protein associated with a transporter.

The engineered *C. thermocellum* strain cannot grow on xylo-oligomer to DCW beyond 0.13 unless a small amount of xylose is also present (Figure 3a) suggesting one of the rate-limiting steps at least in early growth stage could be the lack of sufficient xylanase activity converting complex xylo-oligomer to xylose monomer. We detected

xylobiose as the major hydrolysis product in the spent medium of the engineered *xylAB* strain fermenting xylo-oligomer alone (Figure 3b, middle panel). This observation is consistent with that from Morag et al. (1990) with xylobiose and xylotriose as major hydrolysis products when xylan was incubated for 2 days with a cell-free xylanase prepared from *C. thermocellum*. With prolonged incubation, the authors detected higher amounts of xylose, suggesting xylosidase is a rate-limiting step. To fully realize efficient conversion of complex xylan to high-value products, it is therefore important to boost the endogenous xylosidase activity.

The ability of xylose monomers to potentiate the consumption of xylose oligomers in our engineered *C. thermocellum* strain also indicates that the bacterium can consume xylooligomers directly, but it requires xylose to induce the needed machinery to utilize xylooligomers. Such machinery can include xylanases or the transporters. Similar mechanisms have been observed in cellulolytic fungal species. In the case of *Neurospora crassa*, cellobiose induces the expression of cellulases and hemicellulases, although such induction was masked by the action of β -xylosidase which converts cellobiose to glucose and is also masked by the action of Cre1 which represses cellulases gene expression when a preferred sugar such as glucose is present (Znameroski et al., 2012). In wild-type *C. thermocellum*, the carbohydrate sensing and signaling pathway remains unclear. However, alternative sigma factors and their anti-sigma factors have been suggested to be involved in regulating the cellosomal genes through external carbohydrate-sensing mechanism (Nataf et al., 2010). It is possible that xylose may have triggered the existing carbohydrate sensing/signaling pathways, which elicits regulatory response to enable the consumption of xylooligomers in the *xylAB*-engineered *C. thermocellum* strain.

Both H_2 and ethanol are natural *C. thermocellum* fermentation products. Unlike most hydrocarbons including ethanol, there is no known issue of H_2 toxicity to the microbes during fermentation since H_2 has low solubility in aqueous phase and is released out of the liquid culture. H_2 can serve as an energy carrier to store the excess power during low usage and a clean energy input to grid. It is currently a feedstock for petroleum refining, fertilizer, biofuels and numerous applications. Renewable production of H_2 is thus of broad interests (Meher Kotay & Das, 2008). As shown in our data (Figure 2c), the co-utilization of xylose in addition to cellulose resulted in doubling the production of both H_2 and ethanol. This data projects an increase in the total yield of useful fermentation products per unit (weight) biomass fermented as hemicellulose is also utilized, assuming ~70% of total biomass utilization counting both C5/C6 sugars versus ~40% accounting for only cellulosic sugars. However, our preliminary studies showed that *C. thermocellum* cannot co-ferment xylan and Avicel simultaneously when these two model substrates were fed to the bacteria as the carbon sources (Figure S3, Supporting Information). As shown by this data, xylan was not consumed by the engineered strain in the presence of Avicel. In contrary to the co-fermentation of xylose and Avicel, one explanation is the lack of active xylanases that are essential in depolymerizing xylan when Avicel is present. Hence this data may suggest unknown allosteric or inhibitory effect on certain

xylanase(s) by cellobiose or cellodextrins liberated during cellulose depolymerization. Another possibility is the existence of unknown regulatory mechanisms which down-regulate or repress the gene expression of the genes encoding for essential xylanases in the presence of Avicel, cellodextrins, and/or cellobiose. Our data also indicates that β -xylosidase activity may be limiting in converting xylan to xylose monomers in the given condition (Morag et al., 1990). The potential inhibitory effects at the complex substrate-level remain to be characterized and overcome.

5 | CONCLUSION

The engineered strain presented herein unlocks the possibility toward co-utilizing cellulose and hemicellulose without cross-inhibition in a single strain. Further genetic modifications can be made on this strain to enhance *C. thermocellum*'s ability to utilize mixtures of five-carbon and six-carbon substrates derived from lignocellulosic biomass. Unmasking the cellulose fibers from a lignocellulose matrix via synergistic breakdown of hemicellulose to xylose may further improve the enzyme accessibility to cellulose and efficiency in cellulose hydrolysis. Progress made in this front will lead to shorter and more energetically favorable fermentation processes, and potentially reduce cost associated with pre-processing the plant biomass. The genetic advancement presented herein has immense implications in biorefinery converting lignocellulosic biomass to biofuels and high-value biochemicals.

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